

Pre-fit impact: $\Delta\sigma_{t\bar{t}H}/\sigma^{\text{SM}}, p_T^H \in [120, 200) \text{ GeV}$

$\theta = \hat{\theta} + \Delta\theta$ $\theta = \hat{\theta} - \Delta\theta$

-0.2 -0.1 0 0.1 0.2

Post-fit impact:

$\theta = \hat{\theta} + \Delta\hat{\theta}$ $\theta = \hat{\theta} - \Delta\hat{\theta}$

— Nuis. Param. Pull

$t\bar{t} + \geq 2b$ matching

$t\bar{t}H$ PS & had

$t\bar{t}H$ FSR

JER effective NP 1

$t\bar{t} + 1b$ dipole PS

$k_{t\bar{t} + \text{light}}$ single lep

$t\bar{t} + \geq 2b$ PS H7

b -tag EV 4 b -jets

$t\bar{t} + \geq 1c$ FSR

$t\bar{t} + 1b$ PS H7

$k_{t\bar{t} + \text{light}}$ dilep

$t\bar{t} + \geq 2b$ dipole PS

JER effective NP 2

b -tag EV 6 b -jets

$t\bar{t} + \text{light}$ FSR

$t\bar{t} + \geq 1c$ PS H7

b -tag EV 5 c -jets

$t\bar{t} + \geq 2b$ scale μ_R

Data-driven el fakes alt

b -tag EV 0 light jets

ATLAS

$\sqrt{s} = 13 \text{ TeV}, 140 \text{ fb}^{-1}$

